We now consider a class of differential equations which can be solved explicitly.

Definition. A general first-order differential equation can be written as

$$\frac{dy}{dx} = F(x,y) \tag{1}$$

for some function F of x and y. Such an equation is said to be *separable* if we can write

$$F(x,y) = g(x)f(y)$$

where g depends only on x and f only on y, so that Equation (1) becomes

$$\frac{dy}{dx} = g(x)f(y). \tag{2}$$

Examples.

(a) $\frac{dy}{dx} = xy$ is separable (take $g(x) = x$ and $f(y) = y$).

(b) $\frac{dy}{dx} = \frac{y}{x}$ is separable (take $g(x) = \frac{1}{x}$ and $f(y) = y$).

(c) $\frac{dy}{dx} = \frac{x\cos^2(y)}{1-y^2}$ is separable (take $g(x) = x$ and $f(y) = \frac{\cos^2(y)}{1-y^2}$).

(d) $\frac{dy}{dx} = x+y$ is not separable since $x+y$ cannot be factored in the form $g(x)f(y)$.

Writing Equation (2) as

$$y'(x) = g(x)f(y(x)),$$

we see that, as long as $f \neq 0$, we can solve this equation (at least in principle) by dividing through by $f(y(x))$, integrating both sides with respect to x, and using substitution to transform the left-hand integral:

$$\begin{aligned} \frac{y'(x)}{f(y(x))} &= g(x), \\ \int \frac{y'(x)}{f(y(x))}\,dx &= \int g(x)\,dx \\ \int \frac{dy}{f(y)} &= \int g(x)\,dx \quad \text{(by } u\text{-substitution).} \end{aligned}$$

If the resulting indefinite integrals can be computed explicitly, the result is a relation involving y and x that can (again, in principle) be solved for y.

Example 1. Solve $\dfrac{dy}{dx} = \dfrac{x^2}{y^2}$ subject to the initial condition $y(0) = 2$.

Solution. To find the general solution, we compute the following indefinite integrals:

$$\int y^2\, dy = \int x^2\, dx$$
$$\frac{y^3}{3} = \frac{x^3}{3} + C$$

where C is an arbitrary constant. (We could have used a constant C_1 on the left side and another constant C_2 on the right side. But then we could combine these constants by writing $C = C_2 - C_1$. Since C_1 and C_2 can take any real values, so can C. So, either way, we end up with one arbitrary constant.)

Solving for y, we get

$$y = \sqrt[3]{x^3 + 3C}.$$

Again, since C takes any real value, so does $3C$. It is therefore customary to write the solution as

$$y = \sqrt[3]{x^3 + K} \tag{3}$$

where $K = 3C$ is another arbitrary constant. Equation (3) is the general solution of the differential equation.

To solve the initial-value problem, we now impose the condition $y(0) = 2$, which implies $2 = \sqrt[3]{K}$ and therefore $K = 8$. Thus, the solution of the initial-value problem is

$$y = \sqrt[3]{x^3 + 8}.$$

Example 2. Solve the differential equation $\dfrac{dy}{dx} = \dfrac{6x^2}{2y + \cos(y)}$.

Solution. We find the general solution by calculating the indefinite integrals

$$\int (2y + \cos(y))\, dy = \int 6x^2\, dx$$

which gives

$$y^2 + \sin(y) = 2x^3 + C, \tag{4}$$

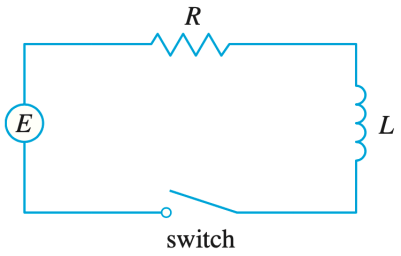
where C is an arbitrary constant. Equation (4) gives the general solution implicitly. In this case, it is impossible to solve the equation to express y explicitly as a function of x.

Exercises

1. Show that the differential equation $y' = x^2 y$ has general solution $y = Ae^{x^3/3}$ where A is any real number.

Name:_____

2. We previously considered the circuit below with $R = 12\Omega$, $L = 4\text{H}$, and $E = 60\text{V}$.



If the switch is turned on at $t = 0$, the current I satisfies the differential equation $\dfrac{dI}{dt} = 15 - 3I$ subject to $I(0) = 0$.

(a) Solve this initial value problem.

(b) Compute $\lim_{t\to\infty} I(t)$.